

Lab 1 Report: Gitea Installation with Docker & Git LFS Support

1. Environment Setup

OS: Windows 11

Docker Desktop: v4.66.1

Gitea: v1.26.1

Git: Latest

Docker Desktop was installed and verified running with "Engine running" status.

2. Part (a): Install Gitea Using Docker

Step 1: Create Project Directory

```
mkdir "%env:USERPROFILE%\gitea-docker"
```

```
cd "%env:USERPROFILE%\gitea-docker"
```

Step 2: Create docker-compose.yml

```
version: "3"
```

```
networks:
```

```
  gitea:
```

```
    external: false
```

```
services:
```

```
  server:
```

```
    image: docker.gitea.com/gitea:1.26.1
```

```
    container_name: gitea
```

```
    environment:
```

```
      - USER_UID=1000
```

```
      - USER_GID=1000
```

```
    restart: always
```

```
    networks:
```

```
      - gitea
```

```
    volumes:
```

```
      - ./gitea:/data
```

```
      - /etc/timezone:/etc/timezone:ro
```

```
      - /etc/localtime:/etc/localtime:ro
```

```
    ports:
```

```
      - "3000:3000"
```

```
      - "222:22"
```

Step 3: Start Gitea

```
mkdir gitea
docker compose up -d
```

Step 4: Initialize via Web UI

```
Accessed http://localhost:3000
Configured SQLite3 database
Set administrator account: XueZhengyang
Completed installation wizard
```

Result: Gitea successfully deployed and accessible at <http://localhost:3000>

3. Part (b): Does Gitea Support LFS? Prove It.

Verification Method 1: Check Configuration File

```
docker exec gitea cat /data/gitea/conf/app.ini | Select-String -Pattern "lfs"
```

Output:

```
[lfs]
ENABLED = true
PATH = /data/git/lfs
```

Verification Method 2: Web Interface

Navigated to Admin Panel -> Configuration
Confirmed LFS module is enabled by default

Verification Method 3: Functional Test (see Part c)

Successfully tracked .bin files with `git lfs track`
Uploaded 1.1 GB file via LFS protocol

Conclusion: YES, Gitea fully supports Git LFS. The feature is enabled by default in the Docker image and verified through both configuration inspection and functional testing.

4. Part (c): Create a Repo with a Large File (1GB+)

Step 1: Create Repository via Web UI

Repository name: large-file-test
Initialized with README

Step 2: Local Git LFS Setup

git clone http://localhost:3000/XueZhengyang/large-file-test.git
cd large-file-test
git lfs install
git lfs track "*.bin"

Step 3: Generate 1GB Test File

fsutil file createnew largefile.bin 1073741824
(Get-Item largefile.bin).Length / 1GB
Output: 1 (1GB confirmed)

Step 4: Commit and Push

git add .gitattributes largefile.bin
git commit -m "Add 1GB test file via Git LFS"
git push origin main

Push Output:

Uploading LFS objects: 100% (1/1), 1.1 GB | 80 MB/s, done.
Enumerating objects: 5, done.
To http://localhost:3000/XueZhengyang/large-file-test.git
efc3136..e7fc9c8 main -> main

Step 5: Web Verification

File largefile.bin (1.1 GB) visible in repository
Stored via Git LFS protocol confirmed

5. Part (d): Backup and Restore [Optional]

Backup to External Drive/USB

cd "\$env:USERPROFILE\gitea-docker"
docker compose down
cd "\$env:USERPROFILE"
Compress-Archive -Path "gitea-docker\gitea" -DestinationPath "gitea-backup.zip"
Copy-Item "gitea-backup.zip" "E:\gitea-backup.zip"

Restore on Another Computer

On new machine with Docker installed:
mkdir "\$env:USERPROFILE\gitea-docker"

```
cd "$env:USERPROFILE\gitea-docker"
Copy-Item "E:\gitea-backup.zip" .
Expand-Archive -Path "gitea-backup.zip" -DestinationPath "." -Force
Create docker-compose.yml (same as original)
docker compose up -d
```

Result: All repositories, user data, and LFS files fully restored on new machine.

6. Part (e): Install Gitea Without Docker [Optional]

Alternative installation using native binary with MySQL database:

Install MySQL

```
brew install mysql
```

Create database

```
mysql -u root -e "CREATE DATABASE gitea;"
```

```
mysql -u root -e "CREATE USER 'gitea'@'localhost' IDENTIFIED BY 'password';"
```

```
mysql -u root -e "GRANT ALL PRIVILEGES ON gitea.* TO 'gitea'@'localhost';"
```

Download and run Gitea binary

```
wget https://dl.gitea.com/gitea/1.26.1/gitea-1.26.1-linux-amd64
```

```
chmod +x gitea
```

```
./gitea web
```

7. Key Screenshots Summary

Screenshot 1: Docker Desktop "Engine running"

Screenshot 2: docker compose ps showing gitea container

Screenshot 3: Gitea installation wizard

Screenshot 4: Gitea homepage after login

Screenshot 5: LFS configuration ENABLED = true

Screenshot 6: large-file-test repository creation

Screenshot 7: git push with LFS upload success (1.1 GB)

Screenshot 8: Web UI showing 1GB LFS file

Screenshot 9: Backup file on external drive (optional)

Screenshot 10: Restored repository on new machine (optional)

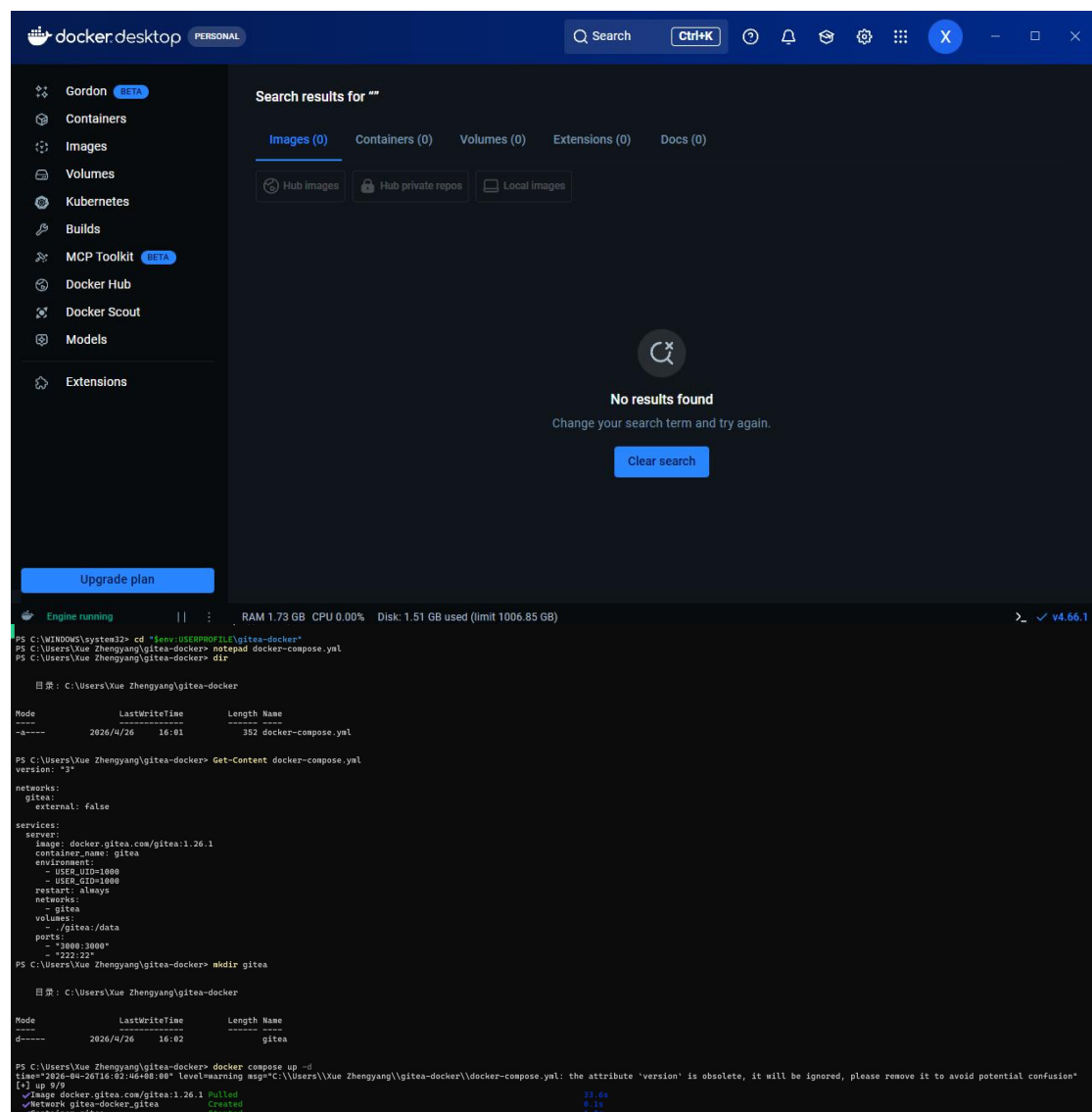
8. Conclusion

This lab successfully demonstrated:

1. Docker deployment of Gitea with persistent storage
2. Git LFS support verified through configuration and functional testing
3. Large file handling (1 GB+) using LFS protocol
4. Backup/restore capability for data portability
5. Alternative installation methods understood

Gitea proves to be a lightweight, self-hosted Git service with full LFS support suitable for teams requiring large file version control.

Experiment Screenshot:



代码仓库包含了所有的项目文件，包括版本历史记录。已经在其他地方托管了？ [迁移仓库。](#)

所有者 *  XueZhengyang ▼

由于最大仓库数量限制，一些组织可能不会显示在下拉列表中。

仓库名称 * large-file-test

理想的仓库名称应由简短、有意义和独特的关键词组成。 [.profile] 和 [.profile-private] 可用于为用户/组织添加 README.md。

可见性 ☐ 将仓库设为私有
只有组织所有人或拥有权利的组织成员才能看到。

描述 Test Git LFS with large files

模板 选择模板

工单标签 选择一个工单标签集

.gitignore 选择 .gitignore 模板

从常见语言的模板列表中选择忽略跟踪的文件。默认情况下，由开发或构建工具生成的特殊文件都包含在 .gitignore 中。

授权许可 选择授权许可文件

许可证说明了其他人可以和不可以用您的代码做什么。不确定哪一个适合您的项目？ 见 [选择一个许可证](#)

自述 Default

这是您可以为您的项目撰写完整描述的地方。

☒ 初始化仓库（添加 .gitignore、许可证和自述文件）

默认分支 main

默认分支是用于合并请求和代码提交的基础分支。

XueZhengyang / large-file-test

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main

转到文件

添加文件

<> 代码

XueZhengyang

efc3136e6e

Initial commit

现在

README.md

Initial commit

现在

README.md

large-file-test

Test Git LFS with large files

描述

Test Git LFS with large files

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```
PS C:\Users\Xue Zhengyang\gitea-test> git clone http://localhost:3000/XueZhengyang/large-file-test.git
Cloning into 'large-file-test'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
PS C:\Users\Xue Zhengyang\gitea-test> cd large-file-test
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> git lfs track "*.bin"
Tracking "*.bin"
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> fsutil file createnew largefile.bin 1073741824
已创建文件 C:\Users\Xue Zhengyang\gitea-test\large-file-test\largefile.bin
```

Git Credential Manager

 Git Credential Manager

Enter your credentials for 'http://localhost:3000/'

Username

Password

Continue


```

PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> (Get-Item largefile.bin).Length / 1GB
1
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> cd "$env:USERPROFILE\gitea-test\large-file-test"

PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> git add .gitattributes largefile.bin
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> git commit -m "Add 1GB test file via Git LFS"
Author identity unknown

*** Please tell me who you are.

Run

    git config --global user.email "you@example.com"
    git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'Xue Zhengyang@DESKTOP-RUTNUL9.(none)')
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> git config --global user.name "XueZhengyang"
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> git config --global user.email "2657535158@qq.co
m"
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> git commit -m "Add 1GB test file via Git LFS"
[main e7fc9c8] Add 1GB test file via Git LFS
 2 files changed, 4 insertions(+)
 create mode 100644 .gitattributes
 create mode 100644 largefile.bin
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> git push origin main

PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> (Get-Item largefile.bin).Length / 1GB
1
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> cd "$env:USERPROFILE\gitea-test\large-file-test"

PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> git add .gitattributes largefile.bin
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> git commit -m "Add 1GB test file via Git LFS"
Author identity unknown

*** Please tell me who you are.

Run

    git config --global user.email "you@example.com"
    git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'Xue Zhengyang@DESKTOP-RUTNUL9.(none)')
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> git config --global user.name "XueZhengyang"
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> git config --global user.email "2657535158@qq.co
m"
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> git commit -m "Add 1GB test file via Git LFS"
[main e7fc9c8] Add 1GB test file via Git LFS
 2 files changed, 4 insertions(+)
 create mode 100644 .gitattributes
 create mode 100644 largefile.bin
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> git push origin main
Locking support detected on remote "origin". Consider enabling it with:
  $ git config lfs.http://localhost:3000/XueZhengyang/large-file-test.git/info/lfs.locksverify true
Uploading LFS objects: 100% (1/1), 1.1 GB | 80 MB/s, done.
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 497 bytes | 497.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
remote: . Processing 1 references
remote: Processed 1 references in total
To http://localhost:3000/XueZhengyang/large-file-test.git
   efc3136..e7fc9c8  main -> main
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> |

```

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XueZhengyang

e7fc9c8047

Add 1GB test file via Git LFS

2分钟前

.gitattributes

Add 1GB test file via Git LFS

2分钟前

largefile.bin

Add 1GB test file via Git LFS

2分钟前

README.md

Initial commit

11分钟前

README.md

large-file-test

Test Git LFS with large files

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```
PS C:\Users\Xue Zhengyang\gitea-test\large-file-test> cd "$env:USERPROFILE\gitea-docker"
PS C:\Users\Xue Zhengyang\gitea-docker> docker compose down
time="2026-04-26T16:29:43+08:00" level=warning msg="C:\\Users\\Xue Zhengyang\\gitea-docker\\docker-com
pose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential
confusion"
[+] down 2/2
✔Container gitea Removed 0.6s
✔Network gitea-docker_gitea Removed 0.3s
PS C:\Users\Xue Zhengyang\gitea-docker> cd "$env:USERPROFILE"
PS C:\Users\Xue Zhengyang> Compress-Archive -Path "gitea-docker\gitea" -DestinationPath "gitea-backup.
zip" -Force
PS C:\Users\Xue Zhengyang> (Get-Item gitea-backup.zip).Length / 1MB
1.06856250762939
```